

Josh Talks: Delivering glitchless online learning experiences with Firebase

Tell us your challenge. We're here to help.

[Contact us \(/contact\)](#)

ABOUT JOSH TALKS

To help its users speaking English in a fun and easy way simultaneously reduce their hesitation, Josh leverages Google Cloud to minimize app response time, and support accurate speech-to-text conversion.



About Josh Talks

Founded in 2015, Josh Talks has evolved from a platform sharing inspiring role model stories, to an organization providing free career guidance and comprehensive online courses in India.

Following this success, Josh Talks launched Josh Skills, an e-learning application featuring interactive English speech exercises tailored for local learners. Since its launch in early 2020, the app now has over 2.5 million downloads.

Industries: Education

Google Cloud results

- Supports real-time language conversation practices with Firebase
- Converts speech to text with more than 90% accuracy rate through Speech-to-Text
- Reduces 30% of operational costs for more investment in product development
- Helps create new interactive language learning features

Short
respo
millise

Location: India

to Text

niveus

About Niveus Solutions

Niveus Solutions is a Google Cloud Partner on a people-centric mission to enable Indian innovators from healthcare, finance, entertainment and more, to optimize their digital transformation. In 2020, it was named Asia Pacific Google Cloud Partner of the Year.

Solutions

Google Cloud (<https://cloud.google.com/>)

Firebase (<https://firebase.google.com/>) Partner

Cloud Speech to Text

(<https://cloud.google.com/speech-to-text/>)

entertainment and more, to optimize their digital transformation. In 2020, it was named Asia Pacific Google Cloud Partner of the Year.

tailored for local users

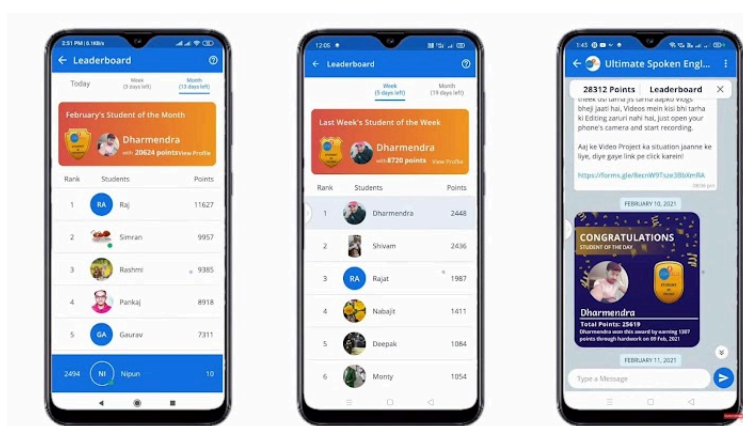
From applications to one-on-one tutoring, there are several resources for people to learn languages online. But when it comes to learning speech, facilitating fun conversations while receiving instant feedback on e-learning platforms might not be easy due to technical limitations like internet disruption and delays in data transfer.

Josh Talks (<https://www.joshtalks.com/>) is dedicated to offering interesting and glitchless online learning experiences. Starting as a platform that shares inspiring role model stories in vernacular languages in India, it has evolved into an organization offering free career guidance and comprehensive online courses over the past seven years. During the onset of COVID-19 in early 2020, Josh Talks launched Josh Skills, which is an e-learning application designed to help users overcome their hesitation of speaking English, which is considered a necessary skill to land decent jobs and to do well in life.

Instead of repeating sentences on the screen, users of Josh Skills can talk about various topics with other learners, while improving their spoken English skills. They can also record their conversations and send it back to the Josh Talks team for feedback on improvement. The application has quickly grown in popularity thanks to these helpful features, and the surge in e-learning products during COVID-19. As of March 2022, Josh Skills has nearly 200,000 paid users and more than 2.5 million downloads.

After running its application for a few months, the Josh Talks team found that there was a slight delay in the communication between its app and end devices, which means that users had to wait for some time before the app responded, to complete their exercises. To offer a more seamless learning experience, Josh Talks migrated its app to Firestore (<https://firebase.google.com/>) on Google Cloud (<https://cloud.google.com/>) in late 2021, because of the low latency that it provides.

"The language learning features of our app are highly interactive, so it is important for us to minimize the app's response time," explains Harsh Gupta, Head of Tech and Product at Josh Talks. "Firestore on Google Cloud stood out for us because of its capability to shorten latency and its high availability."



Delivering glitchless online learning experiences with Firestore

Another challenge involved the notification system that Josh Talks previously used for its app, which

didn't maintain a persistent connection between the app server and end devices, resulting in delays of 8-9 seconds for user notifications. On Firebase, the end devices are subscribed to a collection of data. Whenever the data is updated, app users receive new messages within milliseconds. This way, the users of Josh Talks' app can practice English speech without disruption caused by latency.

"Firebase has helped us greatly improve the user experience and retention rate of our app by minimizing the app response time and realizing real-time language conversation practices," says Gopal Agrawal, Software Development Engineer - III at Josh Talks. "We're now able to provide glitchless learning experiences that enable our users to achieve better results."

"The language learning features of our app are highly interactive, so it is important for us to minimize the app's response time. Firebase on

Google Cloud
stood out for us
because of its
capability to
shorten latency
and its high
availability."

—Harsh Gupta, Head -
Tech and Product, Josh
Talks

Realizing 90% speech recognition accuracy rate with Speech-to-Text

When the users of Josh Talks' spoken English course send recordings of their speech practices for feedback, the Josh Talks team need to turn audio files into text files that can be processed further to detect errors. By leveraging Speech-to-Text (<https://cloud.google.com/speech-to-text>) for transcription, the team is able to realize a speech recognition accuracy rate of over 90%. On top of that, the fact that Speech-to-Text supports Indian English has also helped Josh Talks provide more accurate feedback to its users in India.

"We compared a number of similar products and found that Speech-to-Text has the best transcription accuracy," notes Agrawal. "With Speech-to-Text, we're able to provide feedback to our users more quickly and accurately, which has helped them

sharpen their English speaking skills in a more efficient and effective way."

"Firebase has helped us greatly improve the user experience and retention rate of our app by minimizing the app response time and realizing real-time language conversation practices. We're now able to provide glitchless learning experiences that enable our users to achieve better results."

—Gopal Agrawal,
Software Development

Expanding features to optimize online learning experiences

Josh Talks continues to expand the features of its e-learning app and enhance app performance to deliver even better online learning experiences. The organization is now developing quiz games where users can play in teams on Firestore (<https://cloud.google.com/firestore>), and will use Google Kubernetes Engine (<https://cloud.google.com/kubernetes-engine>) to make its app more scalable by moving it to a microservices architecture. It also plans to employ Natural Language AI (<https://cloud.google.com/natural-language>) to understand the language syntax of its users' speech practices, so that it can provide more helpful feedback.



"Google Cloud
has enabled us
to deliver
significantly
better online
learning
experiences and
increase our
user base
greatly in just a
few months.
Moving forward,
we're confident
that we can
continue helping
our users reach
better learning
results with the
powerful tools of
Google Cloud."

—Gopal Agrawal,
Software Development
Engineer - III, Josh Talks

Agrawal says, "Google Cloud has enabled us to deliver significantly better online learning experiences and increase our user base greatly in just a few months. Moving forward, we're confident that we can continue helping our users achieve better learning results with the powerful tools of Google Cloud."